



SQL Data-Warehouse Project
by [Jakesh Nandurkar](#)



SQL DATA WAREHOUSE

Project



Jakesh Nandurkar

SQL Data-Warehouse Project



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Data Warehousing

"Organize, Structure, Prepare"

- ✓ ETL / ELT Processing
- ✓ Data Architecture
- ✓ Data Integration
- ✓ Data Cleansing
- ✓ Data Load
- ✓ Data Modeling



Exploratory Data Analysis (EDA)

"Understand Data"

- ✓ Basic Queries
- ✓ Data Profiling
- ✓ Simple Aggregations
- ✓ Subquery

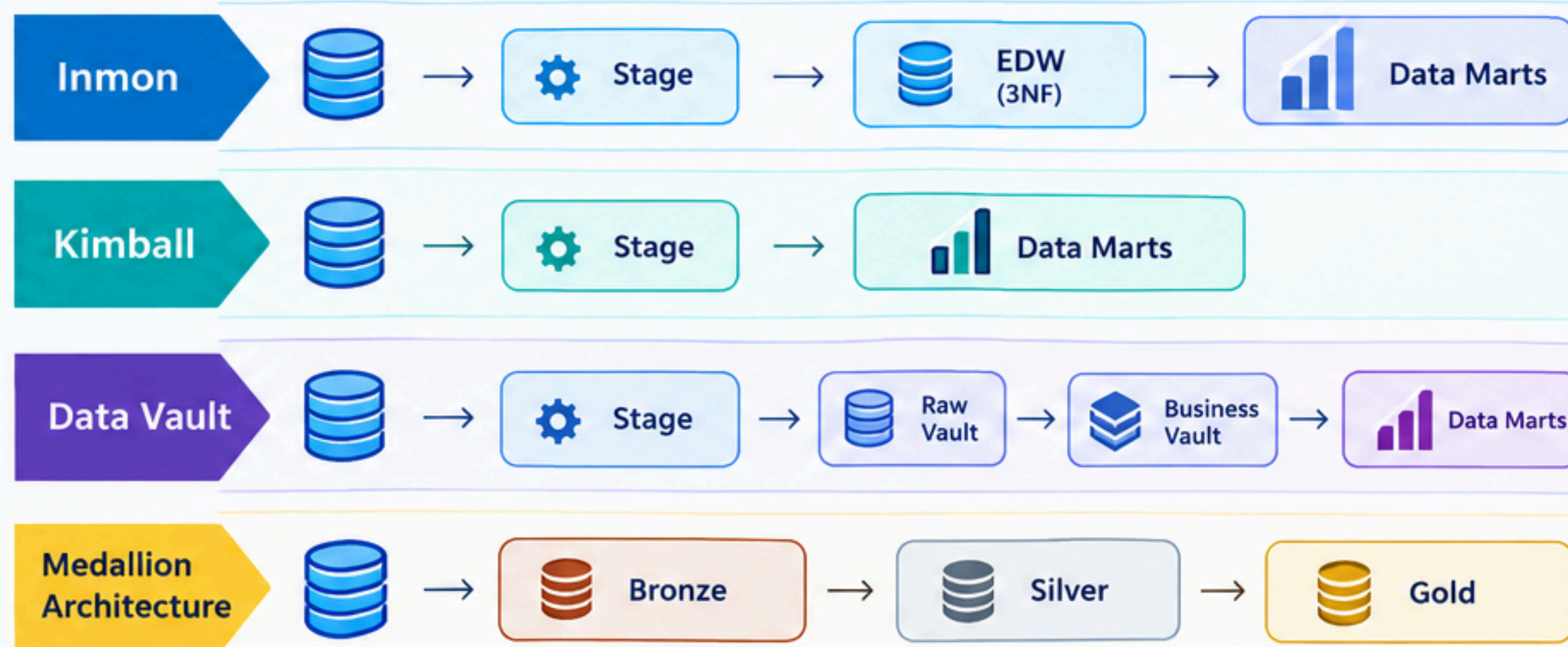
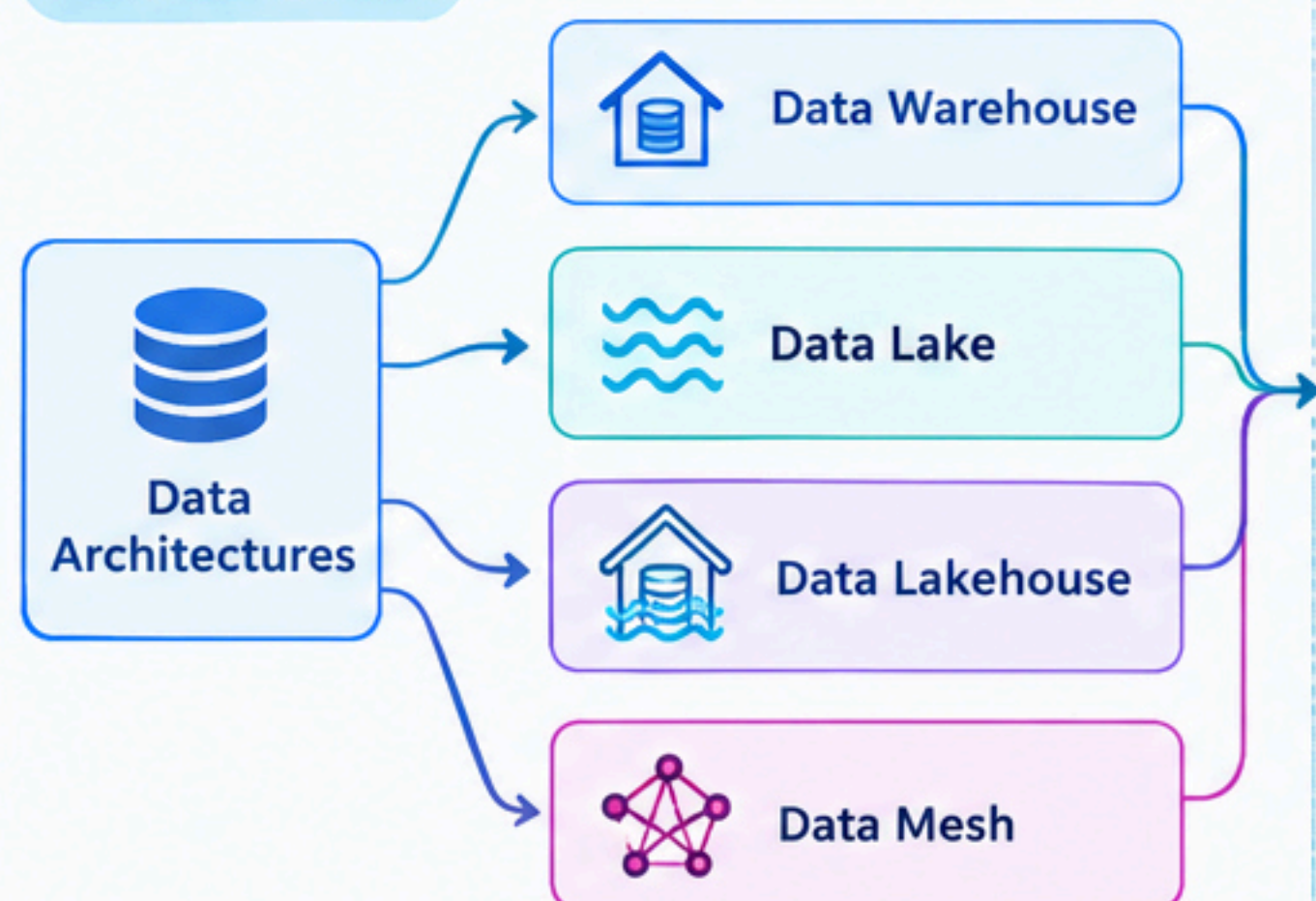


Advanced Data Analytics

"Answer Business Questions"

- ✓ Complex Queries
- ✓ Window Functions
- ✓ CTE
- ✓ Subqueries
- ✓ Reports

Data Sources





Data Warehouse

From Multiple Sources → One Trusted Source → Faster & Smarter Decisions

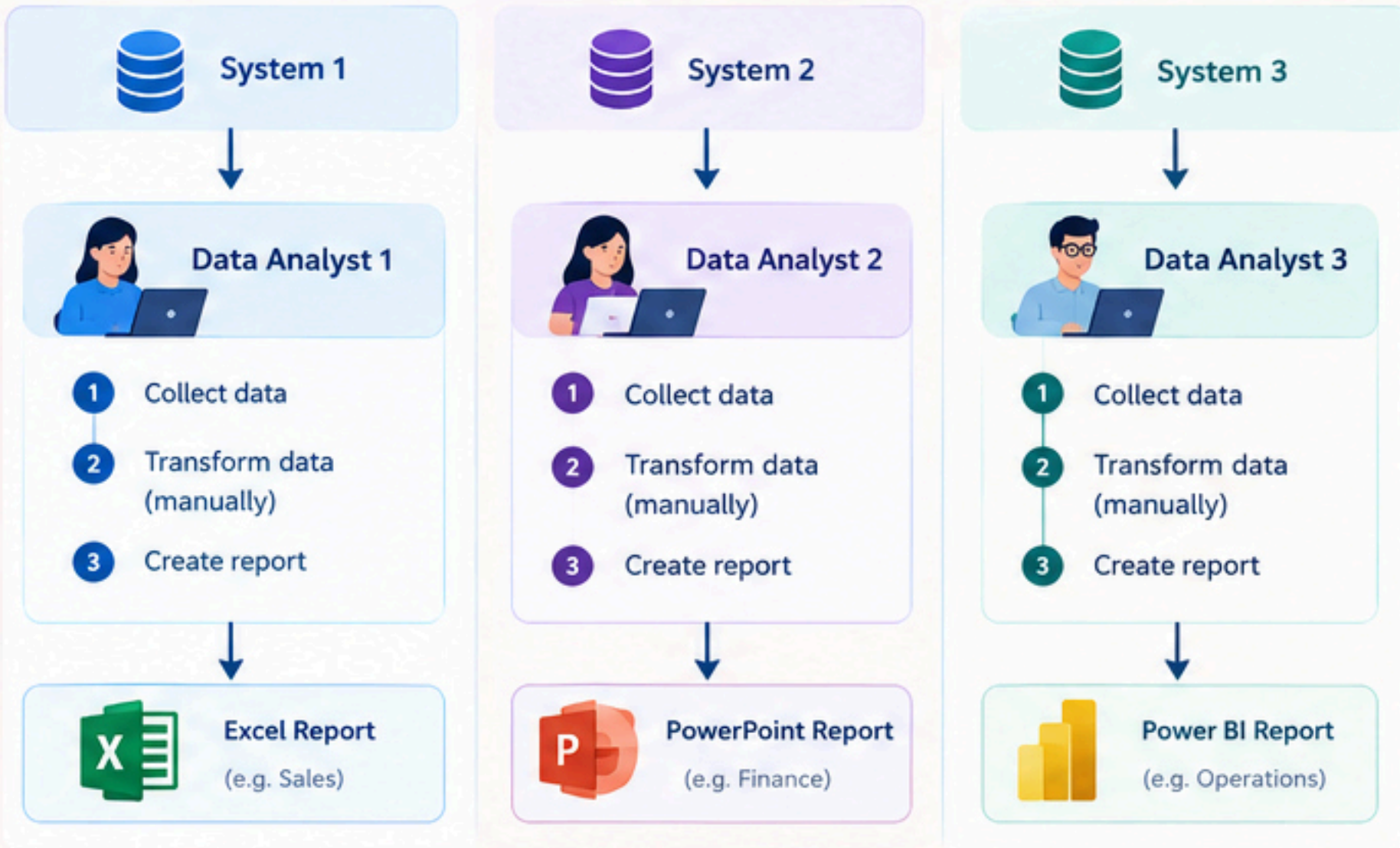
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Without Data Warehouse

Manual • Slow • Error Prone • Inconsistent

Each team collects data from different systems, transforms it manually and creates separate reports. This leads to duplication, delays and confusion.



Problems



Takes too long
(days / weeks / months)



Multiple reports
with different
data versions



Higher chance
of human errors



Hard to handle
large data volumes



Difficult to create
integrated reports
from multiple sources

With
Data Warehouse

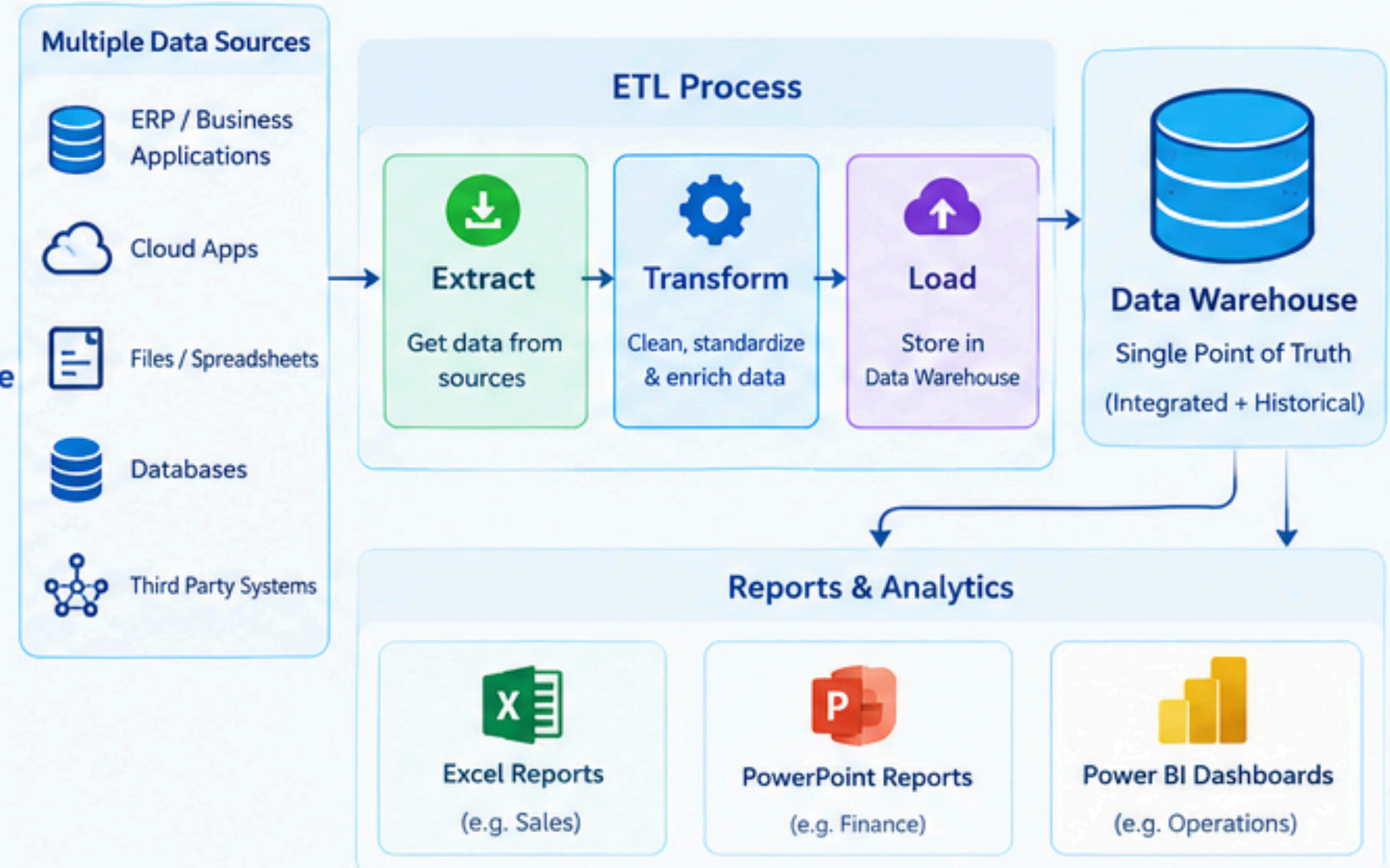
Automated
Integrated
Reliable



With Data Warehouse

Faster • Efficient • Accurate • Unified

Data from multiple sources is extracted, transformed and loaded into a single Data Warehouse. All reports and analytics use the same trusted data.



Data Engineer

- Build ETL pipelines
- Manage data warehouse
- Ensure data quality



Data Analyst

- Explore data
- Build data models
- Create reports & dashboards



Why Companies Build a Data Warehouse?

To bring data from multiple sources into one place, make it easier to analyze, and support better and faster business decisions.

★ Key Benefits



Faster
Reporting



Accurate &
Consistent Data



Integrated
View



Historical
Data Access



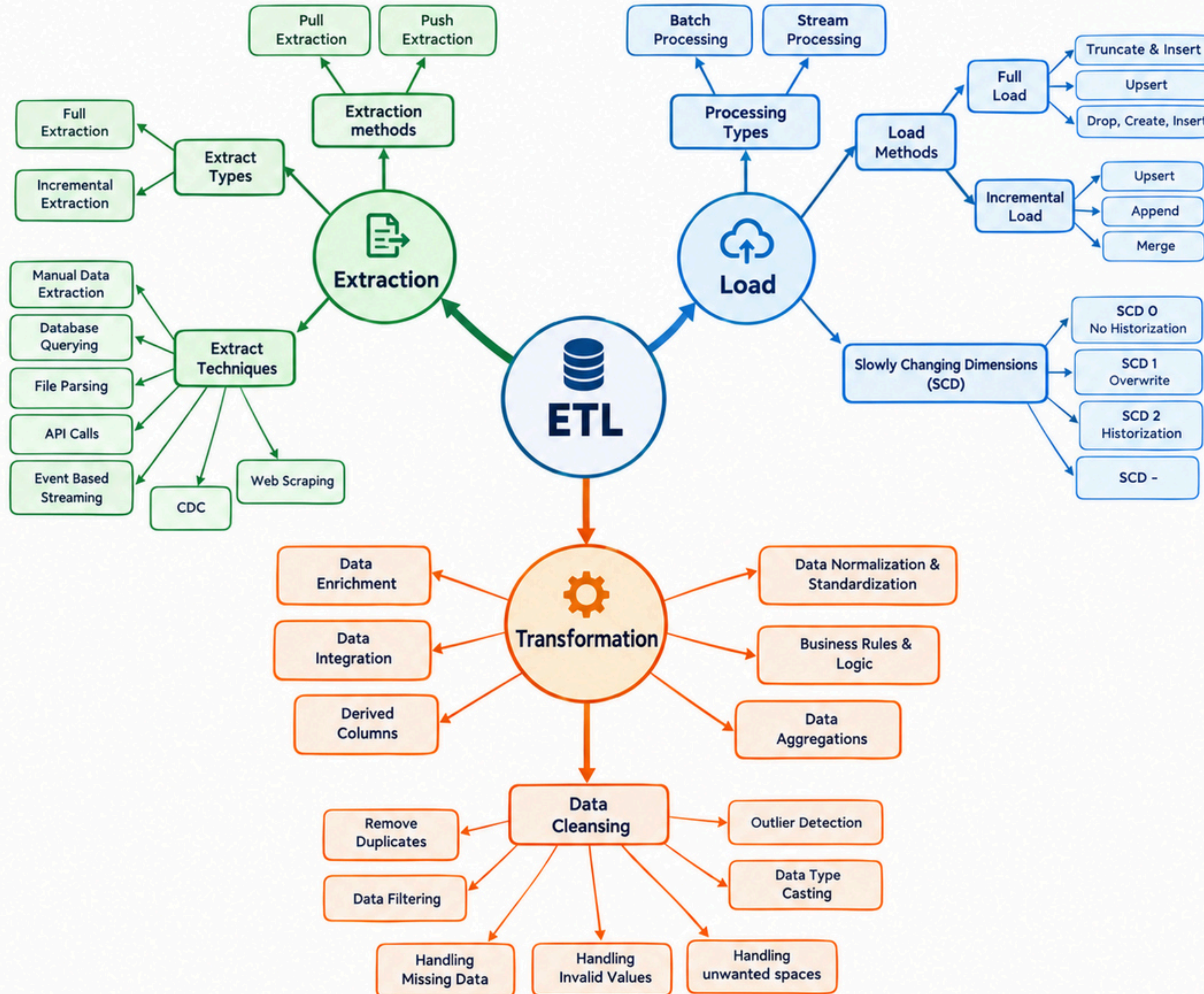
Handles
Big Data



Better
Decision Making

ETL Methods

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**Bronze Layer****Silver Layer****Gold Layer**

	Bronze Layer	Silver Layer	Gold Layer
Definition	Raw, unprocessed data as-is from sources	Clean & standardized data	Business-Ready data
Objective	Traceability & Debugging	(Intermediate Layer) Prepare Data for Analysis	Provide data to be consumed for reporting & Analytics
Object Type	Tables	Tables	Views
Load Method	Full Load (Truncate & Insert)	Full Load (Truncate & Insert)	None
Data Transformation	None (as-is)	- Data Cleaning - Data Standardization - Data Normalization - Derived Columns - Data Enrichment	- Data Integration - Data Aggregation - Business Logic & Rules
Data Modeling	None (as-is)	None (as-is)	- Star Schema - Aggregated Objects - Flat Tables
Target Audience	- Data Engineers	- Data Analysts - Data Engineers	- Data Analysts - Business Users



How a Data Warehouse Works

Restaurant Analogy → From Raw Ingredients to Delicious Results

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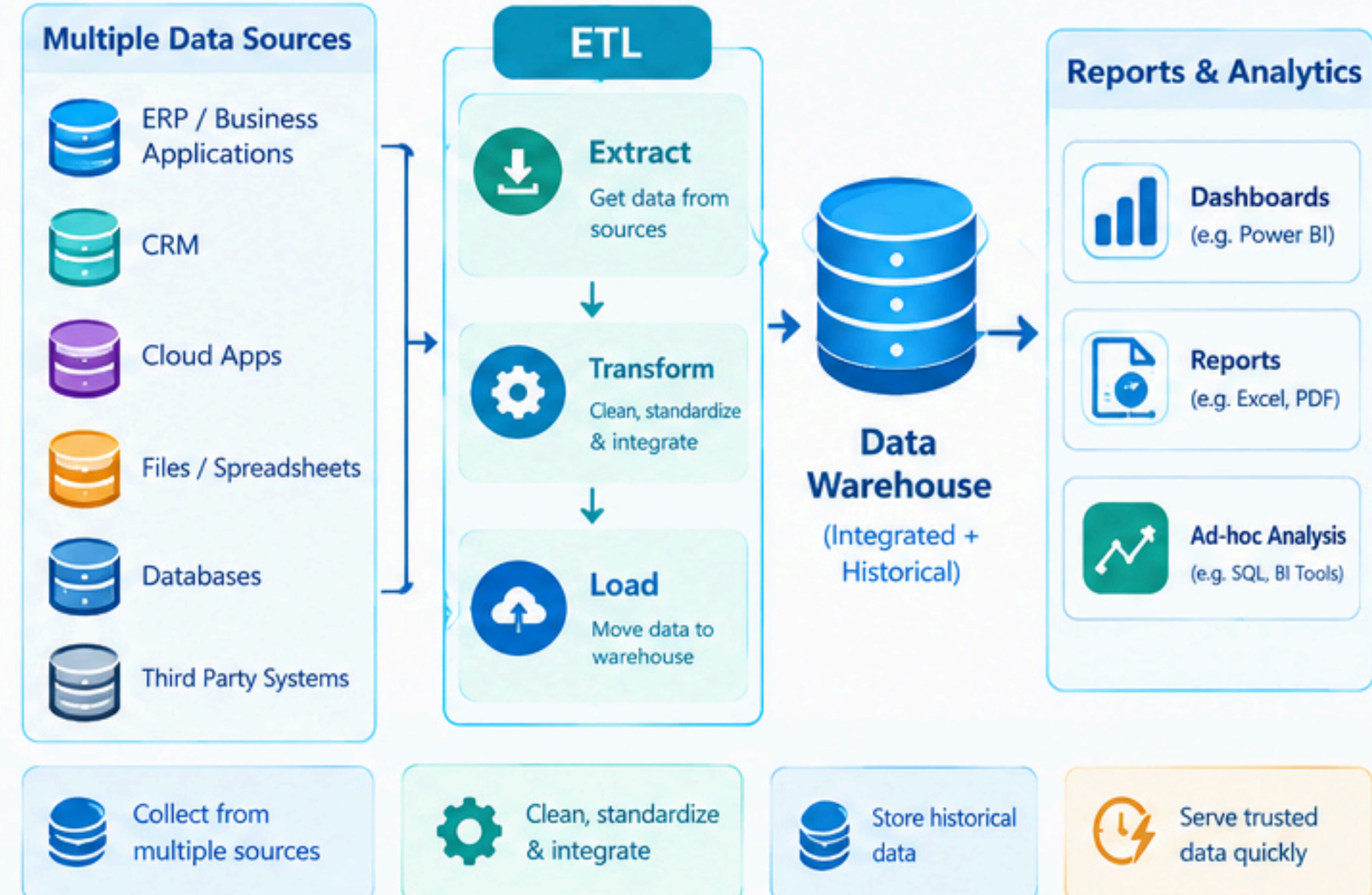
The Restaurant Analogy

Just like a restaurant, a data warehouse collects ingredients from different suppliers, prepares them, stores them in an organized way and finally serves delicious dishes to the customers (business users).



Data Warehouse = The Organized Kitchen

The data warehouse works like a central kitchen where data from multiple sources is collected, cleaned, transformed and stored in one place, then served as trusted reports and insights.



Why It Matters



Faster Reporting

Get insights in hours, not weeks.



Single Source of Truth

One trusted data source for everyone.



Consistent Data

Same data, same results.



Historical Analysis

See trends and make better plans.



Less Manual Work

Automate and reduce human errors.



Better Decisions

Right data, right decisions.



SCD — Slowly Changing Dimensions

How to handle changes in dimension data over time

WHAT IS SCD?

Slowly Changing Dimensions (SCD) is a data-warehouse technique used to manage changes in dimension attributes such as customer city, address or product information.

TYPE 1 — OVERWRITE

- Old value is replaced by the new value.
- No historical version is retained.
- Simple and useful when only the current state matters.

TYPE 2 — KEEP HISTORY

- A new row is created for the changed value.
- Previous versions remain available.
- Useful for historical trend and time-based reporting.

Project note: The reference project focuses on the latest dataset; SCD is shown as a warehouse design concept.



Data Integration

Bringing Multiple Sources Together

MULTIPLE SOURCE SYSTEMS



ERP

Sales • Product



CRM

Customer • Sales



CSV / FILES

Additional data

ETL

1

Extract

2

Transform

3

Load



**Integrated
Data Warehouse**

Single trusted view

WHY INTEGRATE?

1

Consistent

Standardize keys, formats and business definitions.

2

Connected

Combine data across ERP, CRM and files.

3

Trusted

Create one reusable view for reporting.

4

Maintainable

Centralize transformations instead of repeating them.



Data Catalog

Data Dictionary for the Warehouse

WHAT IS A DATA CATALOG?

A data catalog is a centralized repository that stores metadata about the data in your warehouse. It helps users understand what data exists, where it comes from, and how it should be used.

SAMPLE ENTRY IN DATA CATALOG

Table Name	Column Name	Data Type	Description
dim_customer	customer_key	INT	Unique key for customer
dim_customer	customer_name	NVARCHAR	Full name of customer
dim_customer	country	NVARCHAR	Country of customer
fact_sales	sales_key	INT	Unique key for sales
fact_sales	customer_key	INT	Foreign key to customer
fact_sales	sales_amount	DECIMAL	Total sales amount

WHY IT MATTERS



Discover

Find data quickly



Understand

Clarify meaning



Trust

Improve confidence



Share

Support collaboration



End-to-End SQL Data Warehouse Project

Complete project flow from source data to business insights

SOURCE DATA



ERP

Sales • Product



CRM

Customer • Sales



CSV / EXCEL

Additional Data

ETL PROCESS



Extract

Get data from sources



Transform

Clean & standardize



Load

Move data into warehouse

DATA WAREHOUSE (MEDALLION ARCHITECTURE)



Bronze

Raw Data (as-is)



Silver

Clean & Standardized



Gold

Business-Ready

STAR SCHEMA (GOLD LAYER)

Dim Date

Dim Product

Dim Customer

Dim Location

FACT SALES

SQL ANALYTICS & REPORTS

Customer
Insights

Product
Performance

Reports
Excel / Power BI

Business
Insights



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Thank You



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